

From RTL to GDSII: An ASIC Design Course Development using Synopsys® University Program

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Abstract—The development of an ASIC design course using the Synopsys® University Program lectures, labs and tools is presented in this paper. The ASIC design course lasts for 20 weeks and the students learn the design flow using tools including *Design Compiler™*, *IC Compiler™* and *PrimeTime™*. The syllabus is developed based on the Synopsys® University Program Curriculum. Besides the curriculum, the students are assigned projects to synthesize some small circuits to gain more in-depth knowledge about the ASIC design flow. A keystone project is assigned to facilitate the application of the entire IC physical design flow on an industrial size processor design.

Keywords—ASIC design course, Synopsys®

I. INTRODUCTION

Setting up an ASIC design course is a challenging task due to the challenges in tool setup with limited support at universities and the difficulties for students to understand some of the concepts. Significant efforts are necessary in order to provide working computer setups for educational purposes as well as reducing the difficulty levels of the complicated, high variety of ASIC design tool packages for dissemination to the undergraduate and graduate students. In order to help overcome the challenges, Synopsys® University Program [1] provides a set of comprehensive curriculum for the university communities to teach and learn ASIC design using Synopsys® design flow. Moreover, the program also provides a 90nm generic library [2] including standard cell library, I/O library, memories and phase locked loop (PLL) for synthesis. The curriculum includes modular courses, as well as full-length courses at various stages of the design hierarchy, which are used in this paper to create a custom ASIC design flow course.

Based on these resources, an ASIC design course syllabus is developed for the senior undergraduate and graduate students. During the class, the students have the opportunities to learn the basic design flow from RTL to GDSII using *Design Compiler™*, *IC Compiler™* and *PrimeTime™* following the curriculum. The hardware description language (HDL) development to generate the RTL is not targeted in the presented courses, as these topics are covered in other courses in the curriculum, however, can easily be integrated to prelude

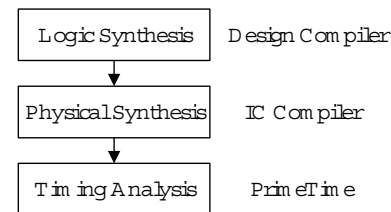


Figure 1. The teaching flow of the ASIC design course.

the ASIC design concepts. Besides following the instructions in the curriculum (lectures and labs) to implement circuits of high complexity, the students are also asked to synthesize a smaller circuit with self-defined objectives such as low power, high performance, etc. This circuit, selected as one of the larger size circuits from ISCAS'89 benchmark circuits, is small enough to visually observe the changes in the physical design but large enough to require implementation through an automation tool. This process greatly helps the students to understand the design flow better and acquire the skills to perform ASIC design. Following a lecture by lecture format to cover these consecutive stages of IC physical design flow and verification, a keystone project is assigned. The keystone project serves to facilitate the application of the entire IC physical synthesis flow on an industrial size processor design.

In Section II of this paper, the course flow is presented. In Section III, the developed course syllabus is proposed. In Section IV, the evaluation of student assessment is presented. The paper is finalized in Section V.

II. THE DESIGN FLOW

The Synopsys® University Program provides a comprehensive set of design tools, library, documentations, curriculum and training videos. In the ASIC design course, the tools and library are setup properly and the curriculum and documentations are selected for the purpose of teaching with reduced difficulty. The teaching flow of the ASIC design course is presented in Figure 1. A significant design decision is to select a subset of the CAD tools, in order to deliver

the fundamental skillset for ASIC design. The proposed ASIC design course focuses on the three stages in Figure 1: Logic synthesis, physical synthesis and timing verification. The discussion of the other IC design flow stages, such as testing, verification, manufacturability, etc. is delegated to other courses.

A. Tools Setup

The tool setup often is a non-critical process, yet can be an issue in the lack of qualified or experienced IT support. The tools are downloaded from the *Solvnet* and installed on a server machine running *Linux Fedora*. A user account is created for each of the students in the class on the server. The user accounts of the students are setup by defining proper environmental variables in each of their *.bashrc* files such that the students can run all the needed *Synopsys*[®] tools. The definition and setup of these environmental variables are similar for most of the *Synopsys*[®] tools and presented in respective tool user guides. The proprietary information potentially requiring non-disclosure agreements (e.g. a third party cell library) is managed through granting access to the user group formed for the class roster.

B. Synopsys University Program Curriculum

Categorized as a full semester courseware, two sets of courses are offered on *solvnet*, targeting *VLSI Design* and *EDA*. The proposed course uses the courseware in the former set. Within the full-semester VLSI Design curriculum courses, those focused on “Digital System Architecture and Design Courseware” are of primary interest. These courses include the following two courses of interest to the authors:

- Basics of Physical Design: Netlist to GDSII,
- Advanced IC Physical Design.

The goal of the first course is to study the contemporary methods of IC physical design and to solve the different problems of physical design with the application of *Synopsys* tools. Some of the main topics covered in this class are listed as “design planning, placement, clock tree synthesis, routing and crosstalk, chip finishing and DFM”. The goal of the second course is to understand the basics of physical design using *IC Compiler*TM. Some of the main topics covered in this class are listed as: “Basic physical design flow using ICC, hierarchical design planning and placement, clock-tree synthesis, routing”. As evident from these descriptions, and easily observed from an examination of the course material, these two courses are very similar and can often be substituted for one another.

C. Design CompilerTM

In order to have the students learn to use *Design Compiler*TM [3], the tutorial and labs are obtained from the “*Synopsys*[®] Implementation Training for Universities” courseware. In this courseware, the first two lectures (Lecture 1 and Lecture 2) and lab assignments (Lab 1 and Lab 2)

are the basics of using the *Design Compiler*TM. In the lab tutorials, the detailed commands for *Design Compiler*TM are listed and some questions are placed to help the students understand the material. During lab session, some guidance from the instructor and the teaching assistant are very helpful due to the relatively steep learning curve of these tools for first time users.

After finishing the first two lectures and lab assignments, the students have a basic understanding of the *Design Compiler*TM. A lab assignment can be given to have the students synthesize a circuit without the provided scripts and setup files. The circuit *s15850* in verilog format selected from ISCAS’89 benchmark circuit set is given to the students as the source design. The 90nm technology library [2] from *Synopsys*[®] is provided as the target library. The students are asked to write a setup file which defines the target library, synthetic library, link library and a *Tcl* script to finish the logic synthesis given some constraints such as timing, power, area information. By finishing these assignments, the students gain a basic skill in performing logic synthesis using *Design Compiler*TM.

Although *Design Compiler Ultra*TM and *Design Compiler Graphical*TM are useful add-ons for the *Design Compiler*TM, their lab sessions are not included in the 20-week ASIC design course. The benefits of the *Design Compiler Ultra*TM and *Design Compiler Graphical*TM are explained at an abstract level, yet the exercises are not covered. Thus the third lab in “*Synopsys*[®] Implementation Training for Universities” curriculum is skipped.

D. IC CompilerTM

In the *IC Compiler*TM [4] classes, the courseware called “Advanced IC Physical Design” is used. Similar to the “*Synopsys*[®] Implementation Training for Universities” courseware, the “Advanced IC Physical Design” courseware provides the lecture slides and the corresponding labs as well. In this curriculum, each step in the physical design flow is presented separately. For instance, floorplanning, placement, clock tree synthesis and routing are discussed in different lecture slides and lab assignments. The students get to learn each step in one week. During the lab session, the students perform the labs provided by this courseware.

After finishing all the lectures and the lab assignments, the students are assigned another homework which is to perform the physical synthesis on the logic synthesized circuit *s15850* output from the *Design Compiler*TM in the previous assignment. Similarly, the students have to write the synthesis script by themselves and perform different optimizations for power, timing, area etc.

This assignment of physical synthesis of an arbitrary RTL design (*s15850.v*), similar to the assignment of the logic synthesis of such an RTL design, is essential. This is such, as the lecture and lab modules used from the *Synopsys*[®]

University Program are rather detailed, which is a double-edged sword. On one hand, these detailed instructions provide a comprehensive, step-by-step instruction set aiding the student learning process of these tools with a steep learning curve and many execution options. On the other hand, these step-by-step instructions sometimes are observed (by the authors) to lead to a hindrance in critical thinking and the overall learning process. The students become too engulfed in completing the steps accurately without getting lost in the maze of options in the CAD tools they are facing for the first time, and might lose sight of the bigger picture.

An additional issue that is addressed with these assignments is the repetition of the learned knowledge in a short period of time. The *Synopsys*[®] *University Program* lab notes follow the IC physical design flow. However, each lab is relatively independent. That is, the placement lab can be performed independent of the floorplanning lab, despite being performed consecutively. Each lab provides an input file that is the completed design of the preceding stage, which is typically the lab that was performed the previous week. This sometimes causes the students to forget information learned in the previous weeks quickly, due to a lack of repetition. The assignments serve to provide an example with minimal assistance (e.g. no step by step instructions) on the synthesis and physical design topics covered in that week's lecture and lab meetings.

E. *PrimeTime*TM

In order to introduce and teach the timing analysis tool *PrimeTime*TM [5], the Lecture 9 slides and lab assignment in the curriculum "Synopsys[®] Implementation Training for Universities" are used. After finishing the lab assignment provided by the curriculum, the students are also asked to do timing analysis on the circuit *s15850*, which helps them to better understand the timing analysis tool. The feedback from the students on the tutorials *PrimeTime*TM is very encouraging. These tutorials are reported to be more comprehensible, following a logical order and focusing on important issues, which were some of the negative feedback on the ICC tutorials discussed in Section II-D.

F. Project

At this stage, the students can use the tools including *Design Compiler*TM, *IC Compiler*TM and *PrimeTime*TM. A project is assigned to the students to synthesize the *OpenSPARC* processor using the curriculum called "Synthesizing a Design Using the 90nm Technology Library". The students learn the synthesis process of a real micro-processor from this project assignment.

Two processor designs – *OpenSPARC* and *PowerPC 405* – are available in the 90nm generic library [2]. However, the *PowerPC 405* files provided are corrupted and thus the *OpenSPARC* is chosen as the project assignment. There is one problem with the *OpenSPARC* as the memory block in

OpenSPARC could not link to the memory library provided. As a result the memory block is compiled to a target library set as the standard cell library [2].

III. SYLLABUS IMPLEMENTATION

Based on the above description, the syllabus for the 20 weeks long ASIC design course is developed as shown in Table I. For the first 13 weeks, weekly assignments include synthesis, placement, routing and verification examples/tutorials using various tools. These tutorials have existing HDL inputs and step-by-step instructions provided by either the *Synopsys*[®] *University Program* or the instructors to learn the tools, which lay the groundwork for a project implementation in the following weeks.

From the 14th week, the keystone design project (*OpenSPARC* synthesis) is assigned. Students are separated into design groups or hierarchical design teams. Laboratory sessions are used to synchronize design teams, similar to weekly progress meetings in an industrial setting. Interface specifications and issues are addressed. Progress reports and presentations are performed. As the student groups are working on the same design project, a competition form can be adopted. The grading scale is established not only to reward the successful completion of the design (which is usually straightforward) but also the performance of the final design. Some design groups are encouraged to target faster operating frequency whereas some others are encouraged for low power implementations. Other variants of optimizations performed by student designer groups include analysis of varying levels of utilization on performance, high-Vt/low-Vt cell swapping for timing improvement and integration with other designs for an SOC implementation.

In academic year 2010-2011 at the Electrical and Computer Engineering department in Drexel University, this course is offered. Drexel ECE operates on a quarter system, where each quarter is approximately 10 weeks long. The proposed 20 week course is divided into two: the first 10 weeks covering the physical design topics lecture by lecture and the second 10 weeks covering timing verification and the keystone design project. The second 10 week course also includes courses on arithmetic circuits, memories etc. to cover in the lectures as the project is assigned in the lab sessions.

IV. ASSESSMENT AND EVALUATION

The assessment of student learning and evaluation is performed summatively. Due to the small number of students in the two consecutive classes (≤ 20) in the academic year 2010-2011, control and tests groups could not be established. Instead, the impact of the course materials (syllabus, lectures, labs and homeworks) from the Synopsys University Program are analyzed within the analysis of the whole course. As such, these impacts are considered to be integral

Table I
THE ASIC DESIGN COURSE SYLLABUS.

Week	Lecture	Lab
1	Introduction to ASIC design	Logic Synthesis setup: "Synopsys® Implementation Training for Universities" Lab 1
2	Logic Synthesis Lecture	Logic Synthesis lab: "Synopsys® Implementation Training for Universities" Lab 2 1 st Assignment (s15850 logic synthesis)
3	Introduction to Physical Design	Intro physical design lab: "Advanced IC Physical Design" Lab 1
4	Floorplanning	Floorplanning lab: "Advanced IC Physical Design" Lab 2
5	Placement	Placement lab: "Advanced IC Physical Design" Lab 3
6	Clock Tree Synthesis	CTS lab: "Advanced IC Physical Design" Lab 4
7	Routing	Routing lab: "Advanced IC Physical Design" Lab 5
8	Multi-mode, Multi-corner Analysis	2 nd Assignment (s15850 physical synthesis)
9	Coping with Interconnect	
10	Project Description, Team and Task Assignments	Timing Lab 1: "Synopsys® Implementation Training for Universities" Lab 9
11	Timing Issues in ASICs	Timing Lab 2: "Synopsys® Implementation Training for Universities" Lab 9
12	Timing Issues in ASICs	Timing Lab 3: "Synopsys® Implementation Training for Universities" Lab 9
13	Timing Verification using Static Timing Analysis	3 rd Assignment (s15850 timing analysis)
14	Timing Verification using Static Timing Analysis	OpenSPARC Project Synthesis + Physical Design
15	Arithmetic Circuits I	
16	Arithmetic Circuits II	OpenSPARC Project Verification + Optimization
17	Progress/Proposal Presentations	
18	Memory Design I	
19	Memory Design II	
20	Final Presentations	

to the course and learning objectives, and the feedback in the student evaluations is scanned for clues about the efficacy of the course materials.

Out of 17 students in both courses, only 6 students completed the survey fully, leading to a response rate of 35.29%. On a scale of 1 to 5, where 1 is "never" and 5 is "always", the students graded the level of involvement of computer technologies to enhance the learning process a 3.8. Despite a standard deviation of 1.1, this was an unexpected result due to the heavy dose of Synopsys labs in this course.

Two interesting results appeared in the student feedback, where the students list the strengths and weaknesses of the instructor and the course. One student wrote:

i would rather want to know what i am going to do in the lab rather than through stuff at us and later explain what we did in the lab because most of the students will not be able to understand the lab and as a result will not be able to complete the labs.

Another student wrote:

There is a disconnect between the lab portion and the lecture portion of the course.

These are strong indicators for the need for instructors to place more emphasis on the synchronization of the material between the lectures and the labs. While the topics in the Synopsys lectures do this on a topic by topic basis, the labs are not always in perfect synchrony with the objectives of the lecture. Two students proposed their solutions:

Utilizing the IC and DC compilers during lectures to demonstrate what is being discussed.

Redo the powerpoint presentations and only include the content you find relevant to the course.

The authors agree to the value of implementing these suggestions at a higher scale.

The second suggestion (above) leads the discussions into the second interesting finding from the evaluation forms. One student, when asked to comment on the key strengths and weaknesses of the instructor(s), wrote:

Very knowledgeable of the course content...but uses powerpoint lectures that are verbatim from Synopsys.

While this might not be a common concern for all students in the class, faculty utilizing these materials should clearly justify the value added to the education process.

V. CONCLUSIONS

The ASIC design class development using the Synopsys® University Program curriculum is presented in this paper. The development of a custom class from multiple courseware is described. Some early evaluation and assessment results are presented, demonstrating the value added and concerns raised in the process.

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